Chapter 1
The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Classification of Compound Propositions

- ***Tautology*** **: compound proposition that is always true**
- ***Contradiction*** **: compound proposition that is always false.**
- ***Contingency*****: compound proposition that is neither a tautology nor a contradiction.**

- **Examples:**

  1) $p \wedge \neg p$ -- **Contradiction**

  2) $p \vee \neg p$ -- **Tautology**

  3) $p \vee \neg q$ -- **Contingency**

Logical Equivalences

【**Definition**】The compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology.

- Notation: $p \equiv q$ or $p \Leftrightarrow q$
- Example: Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

Solution:

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg (p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

De Morgan law

Logical Equivalences

- Show $p \equiv q$

a. Using truth tables

Easy but time-consuming when the number of variables is large

- Construct truth tables for p and q
- Check the truth values of p and q to see if they agree

b. Using already-proved equivalences

- Keep replacing a compound proposition with its equivalent proposition until achieving the goal

Just like what we do in Algebra

Logical Equivalences

Name	Equivalence	
Identity laws	$p \wedge \mathbf{T} \equiv p$	$p \vee \mathbf{F} \equiv p$
Domination laws	$p \vee \mathbf{T} \equiv \mathbf{T}$	$p \wedge \mathbf{F} \equiv \mathbf{F}$
Idempotent laws	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double negation law	$\neg(\neg p) \equiv p$	
Commutative laws	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	
Distributive laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	
De Morgan's laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	

Logical Equivalences

Name	Equivalence
Absorption laws	$p \vee (p \wedge q) \equiv p \quad p \wedge (p \vee q) \equiv p$
Negation laws	$p \vee \neg p \equiv \mathbf{T} \quad p \wedge \neg p \equiv \mathbf{F}$
Implication law	$p \rightarrow q \equiv \neg p \vee q$
Equivalence law	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Logical Equivalences

Examples:

1. **Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.**

Solution:

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q) \quad \text{by the De Morgan's law}$$

$$\equiv \neg p \wedge (\neg\neg p \vee \neg q) \quad \text{by the De Morgan's law}$$

$$\equiv \neg p \wedge (p \vee \neg q) \quad \text{by the double negation law}$$

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \quad \text{by the distributive law}$$

$$\equiv F \vee (\neg p \wedge \neg q) \quad \text{by the negation law}$$

$$\equiv \neg p \wedge \neg q \quad \text{by the identity law}$$

Logical Equivalences

Examples:

2. Show that $((p \to q) \to r) \to ((r \to p) \to (s \to p))$ **is a tautology**

Solution:

$$((p \to q) \to r) \to ((r \to p) \to (s \to p))$$

$$\equiv \neg(\neg(\neg p \vee q) \vee r) \vee (\neg(\neg r \vee p) \vee (\neg s \vee p))$$

$$\equiv ((\neg p \vee q) \wedge \neg r) \vee (r \wedge \neg p) \vee (\neg s \vee p)$$

$$\equiv (\neg p \wedge \neg r) \vee (q \wedge \neg r) \vee (r \wedge \neg p) \vee (\neg s \vee p)$$

$$\equiv (\neg p \wedge (\neg r \vee r)) \vee (q \wedge \neg r) \vee (\neg s \vee p)$$

$$\equiv \neg p \vee (q \wedge \neg r) \vee (\neg s \vee p)$$

$$\equiv T$$

Logical Equivalences

Examples:

3. Show that $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are not logically equivalent.

Hint:

To show these are not logically equivalent, we need only find one assignment of truth values to *p, q, r, s* for which the truth values of the two propositions differ.

Using De Morgan's Laws

- Use De Morgan's laws to express the negation of "Mike has a cellphone and he has a laptop computer" and "Jim will go to the concert or Steve will go to the concert."

Solution:

Mike does not have a cellphone or he does not have a laptop computer.

Jim will not go to the concert and Steve will not go to the concert.

Exercises

- Find a compound proposition involving the propositional variables p, q, r that is true when exactly two of p, q, and r are true and is false otherwise.

Solution:

$$p \wedge q \wedge \neg r$$

$$p \wedge \neg q \wedge r$$

$$\neg p \wedge q \wedge r$$

$$(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$$

Exercises

- The proposition *p NOR q* is true when both *p* and *q* are false, and it is false otherwise. The operator ↓ is called Peirce arrow.

  (a) show that p ↓ p is logically equivalent to ¬ ***p***

  (b) show that (p ↓ q) ↓ (p ↓ q) is logically equivalent to ***p*** ∨ **q**

Propositional Satisfiability

- **A compound proposition is *satisfiable* if there is an assignment of truth values to its variables that make it true. When no such assignments exist, the compound proposition is *unsatisfiable*.**
- **A compound proposition is unsatisfiable if and only if its negation is a tautology.**

Examples:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

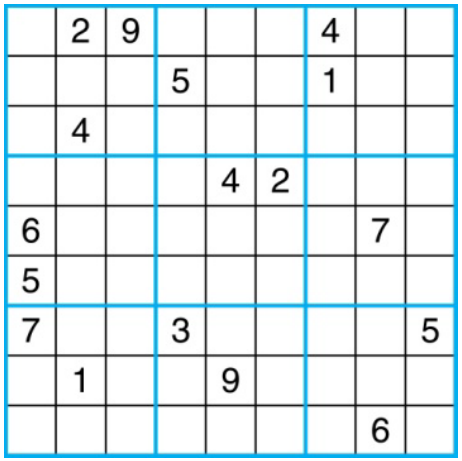
Sudoku

- **A Sudoku puzzle is represented by a 9×9 grid made up of nine 3×3 subgrids, known as blocks. Some of the 81 cells of the puzzle are assigned one of the numbers 1,2, …, 9.**
- **The puzzle is solved by assigning numbers to each blank cell so that every row, column and block contains each of the nine possible numbers.**

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem

- Let $p(i, j, n)$ denote the proposition that is true when the number n is in the cell in the ith row and the jth column.
- There are $9 \times 9 \times 9 = 729$ such propositions.
- In the sample puzzle, $p(5,1,6)$ is true, but $p(5, j, 6)$ is false for $j = 2,3,\ldots 9$



Encoding as a Satisfiability Problem

- **For each cell with a given value, assert *p*(*i*, *j*, *n*), when the cell in row *i* and column *j* has the given value.**
- **Assert that every row contains every number.**

$$\bigwedge_{i=1}^{9} \bigwedge_{n=1}^{9} \bigvee_{j=1}^{9} p(i,j,n)$$

- **Assert that every column contains every number.**

$$\bigwedge_{j=1}^{9} \bigwedge_{n=1}^{9} \bigvee_{i=1}^{9} p(i,j,n)$$

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem

- **Assert that each of the 3 x 3 blocks contain every number.**

$$\bigwedge_{r=0}^{2}\bigwedge_{s=0}^{2}\bigwedge_{n=1}^{9}\bigvee_{i=1}^{3}\bigvee_{j=1}^{3}p(3r+i,3s+j,n)$$

- **Assert that no cell contains more than one number. Take the conjunction over all values of *n*, *n'*, *i*, and j, where each variable ranges from 1 to 9 and $n \neq n'$, of** $p(i,j,n) \rightarrow \neg p(i,j,n')$
- **Find an assignment of truth values to the 729 variables of the form *p(i, j, n)* that makes the conjunction of the above assertions true. Those variables that are assigned T yield a solution to the puzzle.**

Homework

Due date: Mar. 7 (Tuesday)

Ver. 8

Sec. 1.3 6, 8(a,b), 16(b), 36, 44, 55, 62

Ver. 7

Sec. 1.3 6, 8(a,b), 12(b), 32, 40, 51, 58